Case Study

Programme Name & Branch : B.Tech / CSE

Course Name & Code : BCSE302P / Database Systems Lab

Slot : D2+ TD2

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The assignment must be handwritten on A4-size sheets. Typed or printed submissions will not be accepted.

1. Identify all the important concepts represented in the library database case study described below. In particular, identify the abstractions of classification (entity types and relationship types), aggregation, identification, and specialization/generalization. Specify (min, max) cardinality constraints whenever possible. List details that will affect the eventual design but that have no bearing on the conceptual design. List the semantic constraints separately.

Draw an EER diagram of the library database.

Case Study: The Georgia Tech Library (GTL) has approximately 16,000 members, 100,000 titles, and 250,000 volumes (an average of 2.5 copies per book). About 10% of the volumes are out on loan at any one time. The librarians ensure that the books that members want to borrow are available when the members want to borrow them. Also, the librarians must know how many copies of each book are in the library or out on loan at any given time. A catalog of books is available online that lists books by author, title, and subject area. For each title in the library, a book description is kept in the catalog; the description ranges from one sentence to several pages. The reference librarians want to be able to access this description when members request information about a book. Library staff includes chief librarian, departmental associate librarians, reference librarians, check-out staff, and library assistants. Books can be checked out for 21 days. Members are allowed to have only five books out at a time. Members usually return books within three to four weeks. Most members know that they have one week of grace before a notice is sent to them, so they try to return books before the grace period ends. About 5% of the members have to be sent reminders to return books. Most overdue books are returned within a month of the due date. Approximately 5% of the overdue books are either kept or never returned. The most active members of the library are defined as those who borrow books at least ten times during the year. The top 1% of membership does 15% of the borrowing, and the top 10% of the membership does 40% of the borrowing.

About 20% of the members are totally inactive in that they are members who never borrow. To become a member of the library, applicants fill out a form including their SSN, campus and home mailing addresses, and phone numbers. The librarians issue a numbered, machine-readable card with the member's photo on it. This card is good for four years. A month before a card expires, a notice is sent to a member for renewal. Professors at the institute are considered automatic members. When a new faculty member joins the institute, his or her information is pulled from the employee records and a library card is mailed to his or her campus address. Professors are allowed to check out books for three-month intervals and have a two-week grace period. Renewal notices to professors are sent to their campus address. The library does not lend some books, such as reference books, rare books, and maps. The librarians must differentiate between books that can be lent and those that cannot be lent. In addition, the librarians have a list of some books they are interested in acquiring but cannot obtain, such as rare or out of print books and books that were lost or destroyed but have not been replaced. The librarians must have a system that keeps track of books that cannot be lent as well as books that they are interested in acquiring. Some books may have the same title; therefore, the title cannot be used as a means of identification. Every book is identified by its International Standard Book Number (ISBN), a unique international code assigned to all books. Two books

with the same title can have different ISBNs if they are in different languages or have different bindings (hardcover or softcover). Editions of the same book have different ISBNs.
The proposed database system must be designed to keep track of the members, the books, the catalog, and the borrowing activity.

2. Design a database to keep track of information for an art museum. Assume that the following requirements were collected:
   - The museum has a collection of ART_OBJECTS. Each ART_OBJECT has a unique Id_no, an Artist (if known), a Year (when it was created, if known), a Title, and a Description. The art objects are categorized in several ways, as discussed below.
   - ART_OBJECTS are categorized based on their type. There are three main types—PAINTING, SCULPTURE, and STATUE—plus another type called OTHER to accommodate objects that do not fall into one of the three main types.
   - A PAINTING has a Paint_type (oil, watercolor, etc.), material on which it is Drawn_on (paper, canvas, wood, etc.), and Style (modern, abstract, etc.).
   - A SCULPTURE or a statue has a Material from which it was created (wood, stone, etc.), Height, Weight, and Style.
   - An art object in the OTHER category has a Type (print, photo, etc.) and Style.
   - ART_OBJECTs are categorized as either PERMANENT_COLLECTION (objects that are owned by the museum) and BORROWED. Information captured about objects in the PERMANENT_COLLECTION includes Date_acquired, Status (on display, on loan, or stored), and Cost. Information captured about BORROWED objects includes the Collection from which it was borrowed, Date_borrowed, and Date_returned.
   - Information describing the country or culture of Origin (Italian, Egyptian, American, Indian, and so forth) and Epoch (Renaissance, Modern, Ancient, and so forth) is captured for each ART_OBJECT.
   - The museum keeps track of ARTIST information, if known: Name, DateBorn (if known), Date_died (if not living), Country_of_origin, Epoch, Main_style, and Description. The Name is assumed to be unique.
   - Different EXHIBITIONS occur, each having a Name, Start_date, and End_date. EXHIBITIONS are related to all the art objects that were on display during the exhibition.
   - Information is kept on other COLLECTIONS with which the museum interacts; this information includes Name (unique), Type (museum, personal, etc.), Description, Address, Phone, and current Contact_person.

Draw an EER schema diagram for this application. Discuss any assumptions you make, and then justify your EER design choices.

3. Consider a disk with the following characteristics (these are not parameters of any particular disk unit): block size B = 512 bytes; interblock gap size G = 128 bytes; number of blocks per track = 20; number of tracks per surface = 400. A disk pack consists of 15 double-sided disks.
   a. What is the total capacity of a track, and what is its useful capacity (excluding interblock gaps)?
   b. How many cylinders are there?
   c. What are the total capacity and the useful capacity of a cylinder?
   d. What are the total capacity and the useful capacity of a disk pack?
   e. Suppose that the disk drive rotates the disk pack at a speed of 2,400 rpm (revolutions per minute); what are the transfer rate (tr) in bytes/msec and the block transfer time (btt) in msec? What is the average rotational delay (rd) in msec?
   f. Suppose that the average seek time is 30 msec. How much time does it take (on the average) in msec to locate and transfer a single block, given its block address?

g. Calculate the average time it would take to transfer 20 random blocks, and compare this with the time it would take to transfer 20 consecutive blocks using double buffering to save seek time and rotational delay.

4. Which of the following schedules is (conflict) serializable? For each serializable schedule, determine the equivalent serial schedules.
   a. r1(X); r3(X); w1(X); r2(X); w3(X);
   b. r1(X); r3(X); w3(X); w1(X); r2(X);
   c. r3(X); r2(X); w3(X); r1(X); w1(X);
   d. r3(X); r2(X); r1(X); w3(X); w1(X);

5. Consider schedules S3, S4, and S5 below. Determine whether each schedule is strict, cascadeless, recoverable, or nonrecoverable. (Determine the strictest recoverability condition that each schedule satisfies.)
   S3: r1 (X); r2 (Z); r1 (Z); r3 (X); r3 (Y); w1 (X); c1; w3 (Y); c3; r2 (Y);
   w2 (Z); w2 (Y); c2;
   S4: r1 (X); r2 (Z); r1 (Z); r3 (X); r3 (Y); w1 (X); w3 (Y); r2 (Y); w2 (Z);
   w2 (Y); c1; c2; c3;
   S5: r1 (X); r2 (Z); r3 (X); r1 (Z); r2 (Y); r3 (Y); w1 (X); c1; w2 (Z); w3 (Y);
   w2 (Y); c3; c2;

6. A university wants to develop a database system to manage information about students, lecturers, courses, and subjects. Each student is identified by a unique Student ID and has attributes such as Student Name, Date of Birth, Hobby, Age, and Address. The address consists of Door No., Street, City, State, and PIN. A student can attend multiple courses, and each course can be attended by multiple students. Each course has a unique Course ID and Course Name. A course is taught by lecturers, and lecturers may teach multiple courses. Each lecturer has a unique Lecturer ID, Lecturer Name, and Course ID. A course consists of multiple subjects, while each subject belongs to a course. Each subject has a unique Subject ID, Subject Name, and Lecturer ID. Lecturers are assigned to teach subjects.
   Design an Entity–Relationship (ER) diagram for the above University Course Management System.
   Your ER diagram should:
   a. Identify all relevant entities.
   b. Identify the attributes of each entity.
   c. Clearly indicate the primary keys.
   d. Represent the relationships between entities.
   e. Specify the cardinality (1:1, 1:M, M:N) of each relationship.
   f. Represent composite attributes and derived attributes wherever applicable.

7. A large transportation and healthcare management organization is developing a centralized database to manage boats, warehouse inventory, hospital records, and automobiles. During database design, the organization wants to ensure that the relations are free from redundancy and update anomalies.
   As a database designer, analyze the following case studies and normalize each relation up to 3NF wherever required. For each case study:
   - Identify the candidate key.
   - Determine whether the relation is in 0NF, 1NF, 2NF, or 3NF.
   - Identify the relevant functional dependencies.
   - Explain any partial or transitive dependencies.
   - If the relation is not in 3NF, decompose it into appropriate 3NF relations.

Case Study 1 – Ship Management

A shipping company maintains details of its boats, including the number of pools, year of manufacture, tonnage, and the company that owns each boat. A company is uniquely identified by its Company ID, while its name uniquely identifies the company.

Ship(BoatId, NbrOfPools, YearMade, Tonnage, CompanyId, CompanyName)

Business rules:

- BoatId uniquely identifies a boat and determines all its details.
- CompanyId determines CompanyName.
- CompanyName determines CompanyId.

Case Study 2 – Warehouse Inventory Management

A manufacturing organization stores information about parts distributed across multiple warehouses. A part can be stored in several warehouses, and the quantity available depends on both the part and warehouse. Each warehouse has a fixed location.

Inventory(PartNbr, Warehouse, Location, QOH, Weight, PartColor)

Business rules:

- PartNbr determines Weight and PartColor.
- PartNbr + Warehouse determines QOH.
- Warehouse determines Location.

Case Study 3 – Hospital Patient and Test Management

A hospital maintains patient information along with insurance details, doctors, and medical tests performed on each patient. A patient may undergo multiple tests, and each patient-test combination has a corresponding result.

Hospital(Patient, Insurance, Doctor, Test, Result)

Business rules:

- Patient determines Insurance and Doctor.
- Patient + Test determines Result.

Case Study 4 – Automobile Management

- A car dealership maintains a database containing the VIN, manufacturer, style, price, manufacturing year, and color of every car. Each car has a unique VIN.

Cars(Vin, Make, Style, Price, Year, Color)

Business rule:

- VIN uniquely identifies Make, Style, Price, Year, and Color.